

Chapter 6, Problem 76P

Problem

A voltage waveform has the following characteristics: a positive slope of 20 V/s for 5 ms followed by a negative slope of 10 V/s for 10 ms. If the waveform is applied to a differentiator with $R = 50 \text{ k}\Omega$, $C = 10 \text{ }\mu\text{F}$, sketch the output voltage waveform.

Step-by-step solution

Step 1 of 6

Consider the input waveform starts at $t = 0 \text{ ms}$.

For $0 \leq t \leq 5 \text{ ms}$,

$$v_i(t) = 20t$$

For next 10 ms, the slope of the input waveform is 10 V/s.

For $5 \leq t \leq 15 \text{ ms}$,

$$v_i(t) = -10t + c$$

Where c is the intercept on the voltage axis.

Step 2 of 6

Calculate c .

At $t = 15$,

$$0 = -10(15) + c$$

$$c = 150 \text{ V}$$

Thus,

For $5 \leq t \leq 15 \text{ ms}$,

$$v_i(t) = -10t + 150$$

Consider the expression for the input voltage waveform as:

$$v_i(t) = \begin{cases} 20t & 0 < t < 5 \\ -10t + c & 5 < t < 15 \end{cases}$$

Step 3 of 6

Sketch the input voltage waveform as shown in Figure 1.

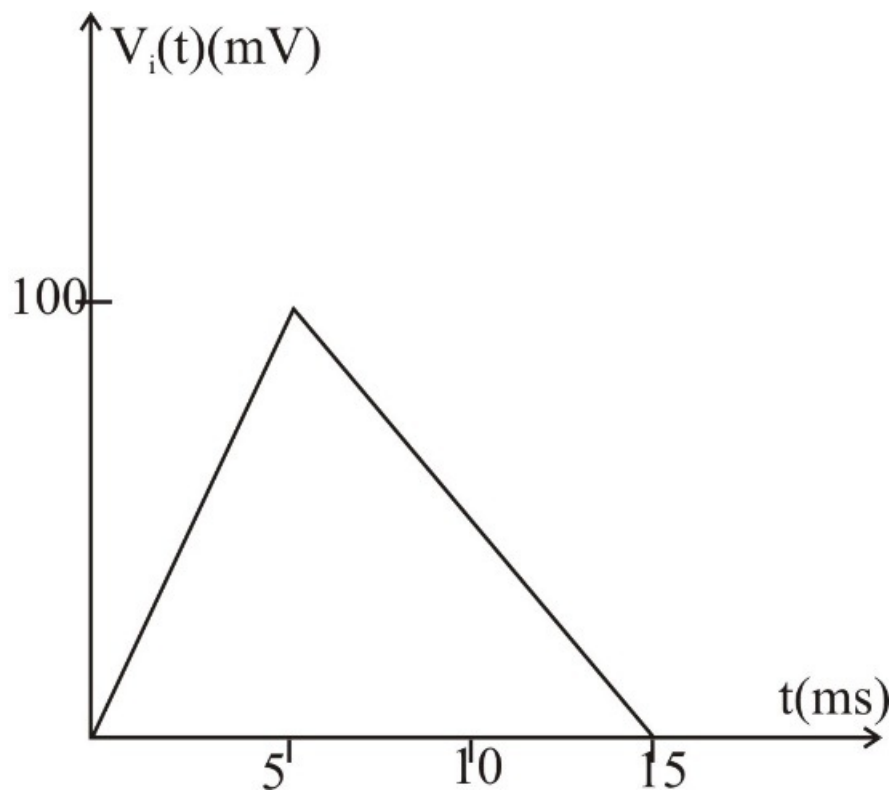


Figure 1

Step 4 of 6

Write the expression for the output voltage, v_o of the differentiator.

$$v_o = -RC \frac{dv_i}{dt}$$

Substitute $50 \text{ k}\Omega$ for R and $10 \text{ }\mu\text{F}$ for C .

$$\begin{aligned} v_o &= -(50 \times 10^3)(10 \times 10^{-6}) \frac{dv_i}{dt} \\ &= -0.5 \frac{dv_i}{dt} \end{aligned}$$

Step 5 of 6

For $0 \leq t \leq 5 \text{ ms}$,

$$\begin{aligned} v_o &= -0.5 \frac{d}{dt}(20t) \\ &= -0.5(20) \\ &= -10 \text{ V} \end{aligned}$$

For $5 \leq t \leq 15 \text{ ms}$,

$$\begin{aligned} v_o &= -0.5 \frac{d}{dt}(-10t + 150) \\ &= -0.5(-10) \\ &= 5 \text{ V} \end{aligned}$$

Step 6 of 6

Sketch the output voltage waveform as shown in Figure 2.

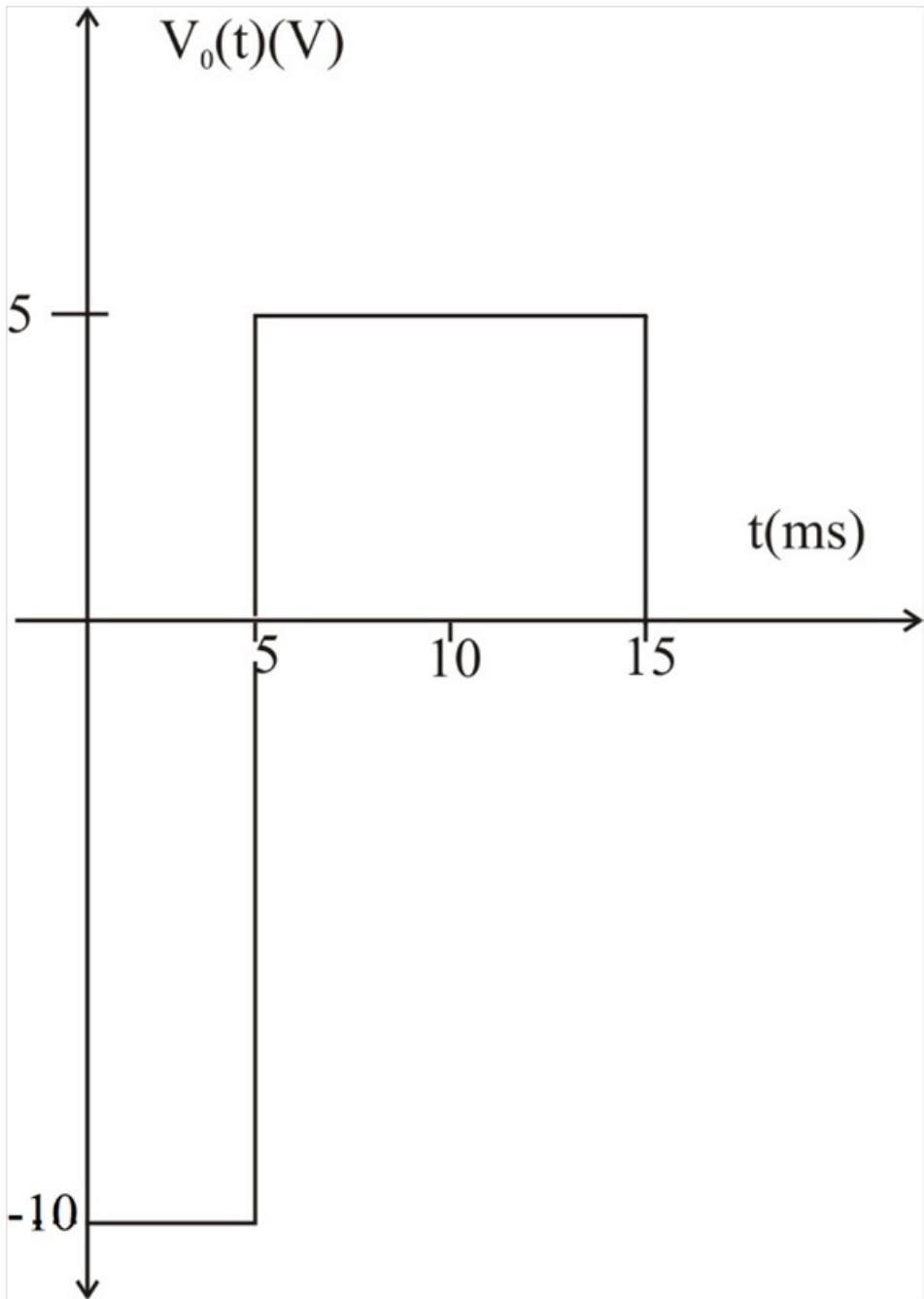


Figure 2